

Cluster Explanations

Authoritative business interpretation of the current five-cluster behavioral segmentation model.

Purpose. This document explains the five behavioral trader segments produced by the current KMeans model. A segment describes observed behavior in the training population; it is not investment advice, a credit or risk rating, or an eligibility decision.

How the model assigns a segment

KMeans groups traders using six features calculated from complete trading history. Every feature is standardized before clustering so measurements with larger numeric units do not dominate distance. The assigned cluster is the nearest of five behavioral centers using the combined standardized feature values. Profile values below are group averages, not membership thresholds.

Feature	Meaning for segmentation
total_trades	Trading activity and frequency.
avg_pnl	Average profit or loss per trade.
roi_pct	Return on investment; compares performance across account sizes.
avg_holding_minutes	Typical holding horizon.
avg_leverage	Typical exposure and aggressiveness.
win_rate	Share of profitable trades.

Current profile reference

ID	Canonical segment name	Traders	Trades	Avg PnL	ROI	Holding	Leverage	Win rate
0	Extreme Risk Traders	10	43.00	-7,421.78	-6.05%	10.25 min	7.42x	24%
1	Conservative Income Traders	51	150.73	472.40	0.05%	2,655.45 min	2.00x	69%
2	High Activity Traders	38	230.71	-882.82	-1.04%	32.50 min	5.02x	43%
3	Structured Strategy Traders	48	92.21	616.35	0.10%	1,015.08 min	2.50x	59%
4	Aggressive Speculators	53	40.60	-4,672.71	-1.85%	11.06 min	7.39x	29%

Population context: 117.31 trades, -0.95% ROI, 930.38 holding minutes, 4.39x leverage, and 49% win rate.

Segment explanations

Trader Intelligence Platform — educational analytics content; not investment advice.

Cluster 0 — Extreme Risk Traders

Definition: the most adverse risk/performance profile: highest leverage, lowest win rate, worst ROI and average PnL, and very short holding periods.

Observed pattern: low-to-moderate trade count, rapid entries and exits, unusually high leverage, and severe losses.

Interpretation: high-exposure short-horizon behavior with severe capital erosion in the training data.

Cluster 1 — Conservative Income Traders

Definition: low leverage, longest holding period, highest win rate, and positive average PnL and ROI.

Observed pattern: moderate activity, multi-session positions, controlled exposure, and consistent profitability.

Interpretation: patient, controlled trading that prioritizes repeatability and risk management.

Cluster 2 — High Activity Traders

Definition: by far the highest trade count and a short holding period, with above-average leverage and moderately negative results.

Observed pattern: frequent intraday trading, quick turnover, and possible cost or slippage exposure.

Interpretation: activity intensity is the defining characteristic; this segment is not synonymous with the most extreme risk.

Cluster 3 — Structured Strategy Traders

Definition: positive average PnL and ROI, good win rate, controlled leverage, and a roughly one-day holding period.

Observed pattern: measured activity, planned positions, controlled exposure, and positive results.

Interpretation: balanced, systematic or structured execution relative to the other groups.

Cluster 4 — Aggressive Speculators

Definition: very high leverage and short holding periods with negative results, but less severe losses than Extreme Risk Traders.

Observed pattern: low trade count, short-horizon positions, high leverage, and below-average hit rate.

Interpretation: aggressive loss-making speculative behavior, distinct from Cluster 0 because outcomes are less severe.

Important comparisons and limitations

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- **Cluster 0 vs. 4:** both are high-leverage and short-horizon; Cluster 0 has materially worse ROI, PnL, and win rate.
- **Cluster 2 vs. 0/4:** Cluster 2 is defined by unusually high trade count; 0 and 4 are primarily defined by leverage and poor outcomes.
- **Cluster 1 vs. 3:** both have controlled leverage and positive outcomes; 1 is more conservative with much longer holding and a higher win rate.
- The training dataset is synthetic. Labels should be revisited after retraining, feature changes, or material differences in production data.

Source basis: project configuration, saved-model cluster profiling, and business definitions. Prepared 12 July 2026.